
Education

- Oct. 2022 - **PhD in Theoretical Neuroscience**, *Group for Neural Theory*, ENS Paris.
 Present
 - *Supervisors*: Alex Cayco Gajic and Jonathan Kadmon. Recipient of a selective scholarship (CDSN)
 - *Title*: Learning dynamics in biologically plausible neural networks: theory and data-driven modeling.
- 2020 - 2022 **M2 MVA (Mathematics, Vision, Learning) & M2 ICFP (Theoretical Physics Track) | ENS Diploma (DENS)**, *ENS Paris & MVA*.
 - *Key Coursework*: Modeling in Neuroscience, Theoretical Neuroscience, Information, inference, networks : from statistical physics to big biological data, Advanced Machine Learning, Information Theory, Algorithms for speech and NLP, Convex Optimization, Quantum Field Theory, Advanced Statistical Physics.
- 2017 - 2019 **Bachelor & M1 in Physics (DENS)**, *ENS Paris*.
 - Admitted through highly competitive national examination (Rank: 25/1137).

Publications

- Sept. 2025 **Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity**, Hugo Ninou, Jonathan Kadmon, N. Alex Cayco-Gajic, **Neurips 2025 Spotlight**.
- March 2025 **Cerebellar output shapes cortical preparatory activity during motor adaptation**, Sharon Israely*, Hugo Ninou*, Ori Rajchert, Lee Elmaleh, Ran Harel, Firas Mawase, Jonathan Kadmon, Yifat Prut, *These authors contributed equally, **Nature Communications**.

Research experience

- Oct. 2022 - **PhD Student**, *ENS Paris*.
 present
 - Developed a biologically inspired learning algorithm and evaluated its efficiency relative to classical gradient descent through large-scale simulations on the university cluster.
 - Conducted statistical analysis of high-dimensional electrophysiological brain recordings to investigate the role of cerebellar signals carrying sensorimotor information in motor preparatory activity during adaptation.
- April 2021 - **Research Intern**, *ENS Paris*.
 July 2021
 - Inference of learning rules in biologically plausible recurrent neural networks via statistical data extraction and data analysis on calcic recordings of zebra fish whole-brain activity.
 - Modeling the brain activity and its learning process based on a Recurrent neural network (RNN) and Generalized linear models (GLM).
- Dec. 2019 - **Research Intern**, *LaDHyX, École Polytechnique - Capital Fund Management*.
 May 2020
 - Data analysis and modeling of emergence phenomena in price heterogeneities.
 - Comparison between usual statistical physics model (Ornstein-Uhlenbeck) to fit the data and conception of an agent based model having phases associated to the different observed behaviors.
- Aug. - Nov. 2019 **Research Intern**, *Canada France Hawaii Telescope*.
 2019
 - Development of a Convolutional Neural Network (CNN) for detecting parasite sources in the data harvested by the SITELLE spectrometer.
 - Conception and training of a deep neural network for computer vision (CNN) able to detect and label the different types of image defects.

Awards

- 2025 **NeurIPS 2025 Scholar Award**.
- 2022 **Awarded a competitive PhD scholarship (CDSN) by the Ecole normale supérieure**.

Teaching experience

- 2018-2019 **Oral examiner (prépa PCSI and MP*)**, *Lycée Michelet and Lycée Saint-Louis*.
- 2024-2025 **Teaching assistant**, *Designed and delivered the tutorials for the course 'Mathematical tools for data science, machine learning, and statistical modeling' taught by Prof. Alex Cayco-Gajic*.

Conferences and presentations

- Mar. 2026 **Friedrich Miescher Institute for Biomedical Research, ZenkeLab**, *Invited talk 'Learning Beyond the Gradient: The Computational Advantage of Adversarial Plasticity'.*
- Mar. 2026 **COSYNE 2026**, *Poster 'Non-Gradient Learning Dynamics with Diverse Neural Populations'.*
- Feb. 2026 **Flatiron Institute**, *Invited talk 'Learning Beyond the Gradient: The Computational Advantage of Adversarial Plasticity'.*
- Dec. 2025 **NeurIPS 2025**, *Spotlight poster 'Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity'.*
- Oct. 2025 **Bernstein 2025**, *Poster 'Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity'.*
- July 2025 **OCNS 2025**, *Poster 'Curl Descent: Non-Gradient Learning Dynamics with Sign-Diverse Plasticity'.*
- Apr. 2025 **ShineLab, University of Sydney**, *Talk on 'Cerebellar output shapes cortical preparatory activity during motor adaptation'.*
- Jan. 2025 **Najafi Lab SYBECO journal club, Georgia Tech**, *Talk on 'Cerebellar output shapes cortical preparatory activity during motor adaptation'.*
- March 2024 **COSYNE 2024**, *Poster 'Cerebellar output shapes cortical preparatory activity during motor adaptation'.*

Other

- Feb. 2025 **Artificial Intelligence Action Summit, Paris**, *Took part in the PSL student assembly as a participant.*